

Masking cannot remove what is inside the region of interest: location-dependent shortcut mitigation across dermoscopy, ultrasound and capsule endoscopy

Abstract

Medical image classifiers exploit acquisition artifacts—hair, sonographer calipers, intestinal debris—that correlate with the diagnosis; the standard defence restricts the model to the region of interest (ROI). This defence depends on where the artifact sits. In pre-registered traps built from real artifacts in four diseases and three modalities, masking removes an out-of-ROI shortcut but leaves an in-ROI one largely intact (crossover +0.15 to +0.38 reversed-test AUROC, all Holm-significant; +0.25 on new patients). Moving the *same* real artifact across the ROI boundary of the *same* images reproduces the effect (location interaction hair +0.25, thyroid +0.28, ovary +0.07, capsule +0.50). It holds for five masking implementations, falls continuously with the artifact–ROI overlap, replicates across encoders, fine-tuning and zero-shot vision–language scores, and grows when images with contradicted labels are removed. A linear-Gaussian account predicted 84 results registered before they existed (mean absolute error 0.011 AUROC). On unaltered data the harm hides in subgroups: overall AUROC barely changes, but masking lowers AUROC on shortcut-conflicting cases, on a patient-disjoint thyroid split and on new patients, and lowers sensitivity among malignant nodules carrying a caliper for a near-benchmark fine-tuned model, in a subgroup registered in advance, partly even after re-tuning the threshold. Against a registered criterion, no remedy dominates masking; the closest, a class-conditional mean constraint on the masked features, removes masking’s failure in every cohort and on every set of conflicting cases, at a cost of 0.02 overall AUROC on one external set. Audits of ROI-masked models should report artifact location and shortcut-conflicting performance. Code, pre-registrations and results are public.

Keywords: shortcut learning, spurious correlation, region-of-interest masking, artifacts, dermoscopy, ultrasound, capsule endoscopy

1. Introduction

Deep classifiers for medical images frequently rely on features that are correlated with the label in the training distribution but are not evidence of disease [9, 7]. In dermoscopy, surgical skin markings [30] and ruler scale bars [31] changed the melanoma predictions of a market-approved system; in chest radiography, pneumothorax classifiers detect the chest drain that treats the condition rather than the condition itself

[20, 14, 33]. Such correlations are causal in origin—suspicious lesions are measured and marked, collapsed lungs are drained—so they are present in every retrospectively collected dataset.

The most widely recommended defence is to restrict the classifier to the region of interest (ROI): segment the lesion or the lungs and discard the rest of the image. The rationale assumes that the shortcut lives in the background. Rulers, hair, ink and drains do not respect that assumption. When the artifact overlaps the ROI, masking cannot remove it; it removes everything *else*. We ask whether the effect of masking depends on where the artifact sits, whether that dependence survives when the artifact is moved while the images stay the same, which quantity governs it, what it costs on unaltered clinical test data, and what to use instead.

Contributions.

1. **A location law for ROI masking.** In pre-registered traps built from real artifacts in four diseases and three imaging modalities, masking removes an out-of-ROI shortcut but leaves an in-ROI one largely intact (Holm-significant crossovers in 4/4 cohorts). Because Trap A and Trap B images differ systematically, we test it on fixed images with a *real-artifact transplant*—the same real artifact pasted inside or outside the ROI of the same images, with a neutral-tissue paste as paste-edge control—and observationally with covariate-matched traps; it replicates across three encoder families, three model sizes, end-to-end fine-tuning in all three cohorts where the network learned the task, and the zero-shot scores of vision–language models; it holds for five masking implementations, is graded in the real overlap, and grows when images with contradicted labels are removed (Secs. 5, 10.2).
2. **Mechanism and a closed-form account.** Masking retains an in-ROI artifact while removing competing evidence, so reliance on it grows; a linear-Gaussian model states when in-ROI masking harms and, fitted to each model’s clean and correlated performance only, matches its held-out reversed-test AUROC far better than simple reference predictors and predicted 84 results registered before they existed (pre-registered; Secs. 6, 7).
3. **Clinical consequences on unaltered data.** Masking leaves overall AUROC nearly unchanged but lowers it on shortcut-conflicting cases, on the patient-disjoint official thyroid split and on new patients (ISIC 2019 → ISIC 2020). For a fine-tuned thyroid model near the published benchmark it lowers sensitivity among malignant nodules that carry a caliper, in a subgroup registered in advance, and part of that loss survives thresholds re-tuned on the test set (Sec. 8).
4. **What to use instead.** A class-conditional mean constraint on the masked features, derived from the law, removes masking’s in-ROI failure in every cohort, including one held out from every choice, and on the shortcut-conflicting cases of every unaltered test set, and is equal to masking elsewhere except for overall AUROC on one external set. Against a registered criterion for dominating masking, tested in two rounds on newly generated data and against published remedies (DFR, JTT, IRM, V-REx,

LEACE, SPLINCE, LaMa inpainting) and a documented search, no remedy dominates; this one fails on one of 42 components. Without artifact labels, an annotation-free mask-then-erase tool handles overlay-like artifacts (Sec. 9; methods in the supplement).

5. **Reproducibility.** Pre-registration of every experiment with a registry of commit times, Holm-corrected primary claims, replication of the original pilot (73/88 numbers), a full rebuild on new hardware that reproduces every primary crossover within 0.01, artifact labels checked without a clinician against public expert annotations and two independent automatic labellers, public code, and a CLAIM 2024 checklist.

2. Related work

Artifacts in dermoscopy. Bissoto et al. [3] annotated seven artifact types in ISIC images and showed that debiasing is harder than it looks; Nauta et al. [19] neutralised coloured calibration patches by inpainting and retraining—an artifact that a colour rule localises and that lies largely outside the lesion. Hair occlusion degrades classifiers trained on HAM10000 [28], and removing dermoscopic dark-corner artifacts is not clearly beneficial, whereas adding synthetic ones to the training set improves classification [22]; both treat an artifact’s presence, not its position relative to the lesion. MaskMedPaint [15] inpaints regions *outside* the classification-relevant area. None of these treats artifact location relative to the ROI as a variable.

Ultrasound. In fetal ultrasound, calipers and on-screen text act as shortcuts even for segmentation [18]; we study them for classification in thyroid and ovarian ultrasound, where they lie on the nodule or tumour.

Chest radiography. Hospital-specific signals limit the generalisation of pneumonia classifiers [33]. Drain-driven pneumothorax shortcuts are documented on NIH-CXR14 and CheXpert [20, 14]; recent work localises them to late layers of probed encoders [21] and mitigates them by prevalence-equalised recalibration [16], which requires the artifact label at test time.

Group robustness and erasure. Reweighting and GroupDRO [24], last-layer retraining [17] and linear concept erasure [23, 1] require artifact labels or paired data; erasure from unpaired labels removes label-correlated signal when the concept is correlated with the label, which is why the mask-then-erase tool we benchmark builds pairs by insertion. Task-preserving linear erasure (SPLINCE; 13) and projection of biased directions in vision–language models [5] are the closest erasure methods; both require concept labels or prompts, whereas our concept subspace is estimated from synthetic insertions. Random erasing and occlusion augmentation [35, 8] use similar overlays as training noise; in our traps this does not remove the shortcut. Test-time selection [2] steers dermoscopy models with user keypoints.

Foundation models. Classifiers trained on frozen foundation-model features inherit shortcut-like behaviour that pooled multi-institutional evaluation can hide [10], and shortcut testing [4] probes whether a model’s performance gaps stem from encoding a sensitive attribute. We ask where in the image the shortcut sits and what masking can and cannot remove.

3. Data

Controlled dermoscopy cohort. ISIC 2018 Task 1/2 images with manual lesion masks and melanoma labels from the union of the ISIC 2016 and 2017 tables (three images with conflicting labels dropped); perceptual-hash (Hamming ≤ 8) and lesion grouping; 2,437 images for which a 65×19 px ruler (0.46% of a 518×518 image) can be placed at 0/25/50/75/100% overlap with the lesion within 10 percentage points (train/val/test 1,849/371/371 before common support).

Real hair cohort (ISIC 2019). All 25,331 ISIC 2019 training images; hair/ruler masks from the archive of Wegley et al. [29]; lesion masks manual for HAM10000 [25] and predicted elsewhere by a U-Net trained on ISIC 2018 Task 1. Two traps share one hair-free group (≤ 30 hair pixels at native resolution): Trap A contains hair with overlap $r = |\text{hair} \cap \text{lesion}|/|\text{hair}| \geq 0.5$, Trap B $r < 0.1$. Cells are subsampled to the hair-free source mix (HAM/BCN/MSK) within each label, giving Trap A 704/157/2,738/989 and Trap B 704/157/4,327/930 images (hair-free benign/melanoma, hair benign/melanoma); five seeds of five-fold group-safe cross-validation (leakage groups: lesion identifiers and pHash); 172 reversed-test melanomas per trap and seed. The 30-pixel threshold is sensitive: independent labels show that it marks hair on many images that experts call hair-free (Sec. 10.2).

Thyroid ultrasound. TN3K images [11] with TNCD benign/malignant labels from cytological biopsy [12] (3,493 images of 2,421 patients; 1,210 malignant) and manual nodule masks; the official train/test split (2,879/614 images) is patient-disjoint [12]. Sonographer calipers were segmented by a frozen rule-based detector (dotted measurement lines and ‘+’ ends; audits in Secs. 10 and 10.2). Trap A: calipers with $r \geq 0.5$ (1,150 images); Trap B: $r < 0.1$ (326); caliper-free: 1,801. Patient identifiers are not published, so within the pooled images leakage groups are pHash chains (stricter embedding-based groups in Sec. 10.3).

Ovarian tumour ultrasound. MMOTU OTU_2d [34] (CC BY 4.0): 1,202 images with tumour masks, labelled benign cystic (chocolate cyst, serous cystadenoma, simple cyst) versus lesions with solid components (teratoma, theca-cell tumour, mucinous cystadenoma, high-grade serous carcinoma). The thyroid detector is applied unchanged: Trap A 341/276, Trap B 91/103, caliper-free 170/178 (benign-cystic/solid).

Capsule endoscopy. SEE-AI frames [32] (CC BY 4.0) containing only erosions (3,550) or only polyp-like lesions (1,931); the ROI is the union of the expert lesion boxes. Contamination (debris, bubbles, bile) is

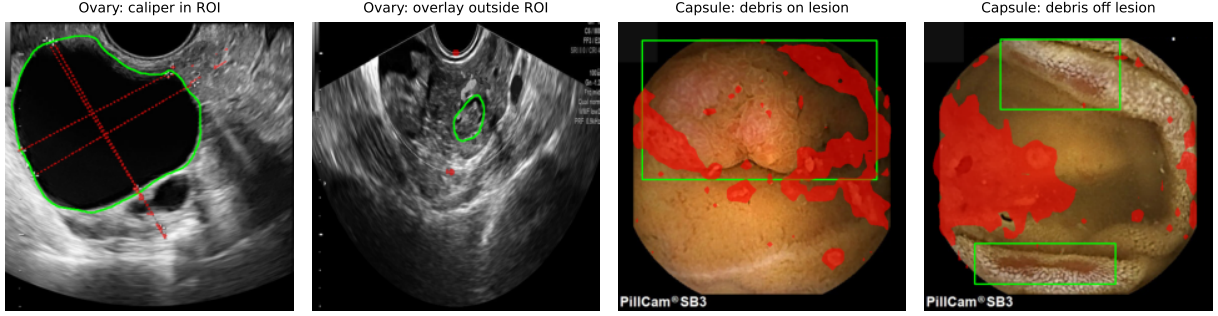


Figure 1: Real in-ROI and out-of-ROI artifacts (openly licensed cohorts). Green: ROI; red: artifact mask. Left: MMOTU ovarian tumours (caliper inside the tumour; overlay outside). Right: SEE-AI capsule frames (debris on and off the lesion box).

segmented by a linear probe on DINOv2 patch tokens trained on 2,160 expert-masked frames (held-out pixel accuracy 0.92, IoU 0.62; expert masks are used where available). Debris-free: $< 3\%$ of the frame; Trap A: debris $\geq 10\%$ with $r \geq 0.5$; Trap B: $r < 0.1$. Leakage groups: blocks of 100 consecutive frame numbers plus pHash duplicates.

Chest radiography. NIH ChestX-ray14 [27]; lung ROIs from the TorchXRyVision PSPNet [6]. *Controlled cohort:* 1,906 pneumothorax images without a chest drain (NEATX annotations) and 5,718 *No Finding* images from other patients; a 100×12 px radio-opaque tube is placed at controlled overlap with the lungs (7,610 images in common support). *Real drains* (in-ROI only): 29,687 images; drains are labelled by NEATX for pneumothorax-positive images and by a drain detector (precision 0.95) for negatives; environments are built within view position (AP/PA); five patient-grouped folds.

4. Problem setup and protocol

Let $Y \in \{0, 1\}$ be the diagnosis and $A \in \{0, 1\}$ the presence of an artifact. A *trap* trains with $P(A=1 | Y=1) = 0.9$ and $P(A=1 | Y=0) = 0.1$ and tests on a correlated environment with the same rates, a *reversed* environment with the rates swapped (0.1/0.9), and a clean environment in which the artifact is independent of the label (real traps: present in half of each class; controlled sweeps: absent). A model that uses A gains on the correlated test and collapses on the reversed one; the primary metric is reversed-test AUROC, and clean AUROC measures the cost of a mitigation. The overlap $r = |\text{artifact} \cap \text{ROI}| / |\text{artifact}|$ is the controlled variable. Three designs vary it: *controlled sweeps* paste a synthetic artifact at overlaps 0–100% on artifact-free images; *real traps* compare images whose real artifact lies inside the ROI ($r \geq 0.5$, Trap A) or outside it ($r < 0.1$, Trap B), each added to one shared artifact-free group; the *real-artifact transplant* pastes one real artifact instance, cut out with its mask, inside ($r \geq 0.95$) or outside ($r = 0$) the ROI of the same artifact-free images. The *location crossover* $[\text{mask} - \text{ERM}]_{\text{out}} - [\text{mask} - \text{ERM}]_{\text{in}}$ of reversed-test AUROC summarises

each design. Unless stated otherwise backbones are frozen and linear heads are fitted (end-to-end fine-tuning in Sec. 5.6); regularisation strength and the decision threshold (balanced accuracy) are selected on clean validation data only. Uncertainty is a paired *crossed* bootstrap: training seeds (or cross-validation folds) are resampled as clusters, each test image receives one Poisson weight shared by every cluster in which it appears, and AUROC differences are computed inside each cluster, so predictions of different models are never pooled into one ROC curve (10,000 replicates unless stated). The original protocol resampled images independently within each seed, which treats a test set shared by the seeds as several and gave intervals a median 1.56 times narrower (IQR 1.34–1.74 over 2,466 intervals; Supplement S15). Every controlled sweep, real trap, transplant, remedy and unaltered-data analysis was therefore regenerated from per-image predictions, and *every interval of a model comparison in the paper and the supplement is a crossed-bootstrap interval*; none uses the per-seed estimator. Two further cases are stated where they occur: agreement statistics between label sources, which involve no training seeds, use an image bootstrap; and a few secondary analyses in the supplement (patch-token suppression, pseudo-group balancing, erasure combined with DFR, the validation-driven selectors, text-prompted erasure, SPLINCE on chest drains and LaMa on dermoscopic hair) come from archived runs whose per-image predictions were not kept and are reported as point estimates, marked as archived. Registration status: the real traps, sweeps and unaltered-data tests were registered before their results (the 16-test primary family was grouped afterwards); the covariate-matched traps, the transplant, its paste-edge control and the operating points were designed after the real-trap results and registered before their own results, and each forms its own Holm family outside the primary family (Supplement S1). The analyses added later—masking implementations, overlap dose-response, the external natural test, the prospective theory test, the label checks, the fine-tuned thyroid model and the threshold re-tuning—were registered in four further documents, each committed and pushed before its analysis; analyses decided after their data were seen are labelled exploratory. Splits are grouped by lesion/patient identifier where published and otherwise by perceptual hash (and frame blocks); a stricter embedding-based grouping is a sensitivity analysis (Sec. 10.3).

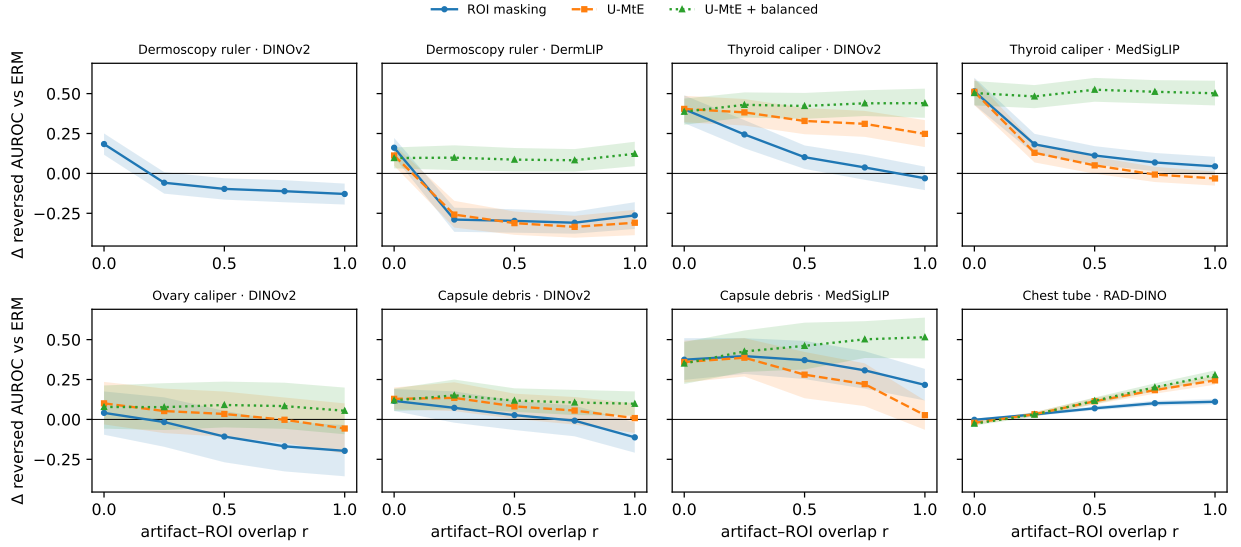
5. Results: a location law for ROI masking

5.1. Controlled dermoscopy: the effect reverses with overlap

With a synthetic ruler whose overlap with the lesion is swept from 0 to 100% (2,437 ISIC 2018 images, DINOv2 ViT-B/14 at 518 px, three seeds), ROI masking improves reversed-environment AUROC over ERM by +0.184 [+0.117, +0.250] when the ruler lies outside the lesion and *lowers* it by −0.097 [−0.164, −0.030], −0.111 [−0.181, −0.043] and −0.129 [−0.195, −0.064] at 50, 75 and 100% overlap (location interaction +0.313 [+0.258, +0.370]; Table 1); at 25% overlap the change is not significant (−0.059 [−0.127, +0.008]). The sign flip is reproduced at 224 px and under randomised ruler colour, opacity, ticks and scale. At 0% overlap the

Table 1: Synthetic ruler, ISIC 2018, DINOv2 ViT-B/14 @518: AUROC and reversed-test Δ vs ERM [95% CI]

Method	Needs	clean@0	rev@0	clean@50	rev@50	clean@100	rev@100	Δ rev@0	Δ rev@50	Δ rev@100
ERM	—	0.797	0.665	0.793	0.573	0.792	0.515			
ROI masking	ROI mask (train+test)	0.849	0.849	0.829	0.475	0.825	0.386	+0.184 [+0.117, +0.250]	-0.097 [-0.164, -0.030]	-0.129 [-0.195, -0.064]
Inpainting (oracle)	artifact pixels (train+test)	0.810	0.806	0.809	0.783	0.807	0.758	+0.141 [+0.109, +0.175]	+0.211 [+0.175, +0.249]	+0.243 [+0.205, +0.284]
Group-balanced	image-level A (train)	0.814	0.802	0.815	0.791	0.806	0.796	+0.137 [+0.099, +0.179]	+0.219 [+0.177, +0.265]	+0.281 [+0.236, +0.329]
DFR	image-level A (val)	0.786	0.789	0.787	0.794	0.789	0.794	+0.124 [+0.075, +0.179]	+0.221 [+0.166, +0.281]	+0.279 [+0.220, +0.339]
LEACE (paired, oracle renders)	paired renders (train)	0.829	0.832	0.810	0.810	0.809	0.809	+0.167 [+0.126, +0.211]	+0.237 [+0.197, +0.280]	+0.294 [+0.249, +0.339]
Repair consistency	artifact pixels (train)	0.828	0.709	0.822	0.616	0.814	0.557	+0.044 [-0.002, +0.091]	+0.044 [-0.007, +0.092]	+0.042 [-0.009, +0.092]
Insert-then-erase (artifact templates)	artifact templates	0.849	0.844	0.850	0.833	0.847	0.818	+0.179 [+0.136, +0.225]	+0.260 [+0.212, +0.311]	+0.303 [+0.250, +0.357]
Insert-then-erase + balanced (artifact templates)	artifact templates + A (train)	0.841	0.842	0.843	0.840	0.844	0.842	+0.177 [+0.135, +0.221]	+0.267 [+0.219, +0.316]	+0.327 [+0.276, +0.380]
Insert-then-erase rank-1 (ablation)	artifact templates	0.811	0.799	0.809	0.770	0.806	0.730	+0.134 [+0.104, +0.166]	+0.197 [+0.161, +0.235]	+0.215 [+0.178, +0.254]
Insertion augmentation	artifact templates	0.814	0.726	0.812	0.653	0.810	0.603	+0.061 [+0.045, +0.079]	+0.081 [+0.064, +0.099]	+0.088 [+0.072, +0.105]


 Figure 2: Controlled overlap sweeps: change in reversed-test AUROC versus ERM as the synthetic artifact moves into the ROI (95% CIs shaded). ROI masking (blue) declines with overlap in every modality except chest radiography, where the encoder ignores extra-pulmonary tubes; U-MtE with group balancing (green) is flat. Generated by `scripts/make_figures.py`.

masked model’s correlated and reversed AUROC coincide (0.849/0.849); at 100% its correlated AUROC exceeds ERM’s (0.961 vs 0.937): it uses the ruler *more*. Synthetic calipers on thyroid nodules, synthetic debris on capsule lesions and synthetic calipers on ovarian tumours give the same interaction (thyroid +0.433 [+0.362, +0.507], capsule +0.228 [+0.144, +0.324], ovary +0.238 [+0.144, +0.340]; Fig. 2); whether the lost benefit becomes harm depends on the setting (ovary at full overlap $-0.197 [-0.357, -0.035]$; thyroid $-0.030 [-0.105, +0.042]$).

5.2. Real artifacts in four diseases (pre-registered)

Each real trap pair shares one artifact-free group; Trap A adds images whose artifact lies mostly inside the ROI ($r \geq 0.5$), Trap B images whose artifact lies outside it ($r < 0.1$); five seeds of five-fold group-safe cross-validation. The crossover $[\text{mask} - \text{ERM}]_{\text{Trap B}} - [\text{mask} - \text{ERM}]_{\text{Trap A}}$ is positive and Holm-significant in all four cohorts: dermoscopic hair +0.147 [+0.111, +0.185], thyroid calipers +0.239 [+0.196, +0.280], ovarian

calipers +0.170 [+0.087, +0.252] and capsule debris +0.377 [+0.327, +0.426]. On dermoscopy masking *harms* in-ROI (−0.049 [−0.083, −0.018], negative in all five seeds; BCN −0.067 [−0.110, −0.025]) and helps out of the ROI (+0.098 [+0.060, +0.136]). On thyroid, ovarian and capsule images in-ROI masking still helps, but far less than out of the ROI (thyroid +0.109 [+0.085, +0.133] versus +0.348 [+0.307, +0.388]; capsule +0.084 versus +0.461), and the masked Trap A model still relies on the calipers (correlated 0.908 versus reversed 0.398). On new patients — ISIC 2020, 32,997 images of 2,056 patients, patient-disjoint folds, hair masks from a segmenter trained only on ISIC 2019 and frozen before any ISIC 2020 image was scored — the pre-registered external test reproduces the hair pattern: masking harms with hair on the lesion (−0.064 [−0.128, −0.002]), helps with hair beside it (+0.186 [+0.118, +0.249]), crossover +0.250 [+0.174, +0.327]. Because the artifact-bearing images of the two traps differ (next subsections), these crossovers describe the traps as built; their attribution to location rests on the matched traps for hair, thyroid and ovary and on the transplant for capsule, where matching is infeasible. What is universal in our data is therefore the *gap*: masking leaves an in-ROI shortcut largely intact while it removes an out-of-ROI one. Harm is conditional; it appears when the mask also removes disease context or leaves the artifact a larger share of what remains (Sec. 7).

5.3. Same images, same artifact: real-artifact transplant and paste-edge control

Trap A and Trap B contain different artifact-bearing images, and these differ systematically (Supplement S11): lesions carrying in-lesion hair are much larger (standardised mean difference 2.1), as are thyroid nodules with calipers on them (0.4) and capsule lesions with debris on them (2.6), and in-ROI calipers carry more marker pixels. The crossover could therefore reflect population differences rather than artifact location. We removed this possibility by design (pre-registered): real artifact instances—a caliper measurement line with its end crosses, a hair strand, a piece of debris—were cut from artifact-bearing images with their masks and pasted *inside* (overlap ≥ 0.95) or *outside* (overlap 0) the ROI of the *same* artifact-free images, with the same instance, transform and compositing in both locations; environments as in the controlled sweeps. As a paste-edge control, neutral tissue was pasted through the same masks, at the same positions and with the same blending (Supplement S11). With the artifact outside the ROI masking removes the shortcut entirely—the masked model’s reversed, correlated and clean AUROC coincide—whereas with the identical artifact inside the ROI the shortcut survives masking (Supplement S11). The location interaction is positive in all four cohorts: dermoscopic hair +0.253 [+0.214, +0.293], thyroid +0.277 [+0.256, +0.298], ovary +0.074 [+0.044, +0.104], capsule +0.497 [+0.427, +0.567] (Table 2). *Paste-edge control (pre-registered)*. Pasting neutral tissue through the same masks, at the same positions and with the same blending gives a smaller but still positive interaction (dermoscopic hair +0.204 [+0.164, +0.244], thyroid +0.145 [+0.130, +0.161], ovary +0.052 [+0.021, +0.084], capsule +0.459 [+0.383, +0.535]), 52–92% of the artifact’s. The neutral patch is itself a local, label-correlated pattern that the encoder learns, and masking removes it only outside the ROI.

Table 2: The location law under four designs (reversed-test AUROC, DINOv2 ViT-B/14; 95% crossed seed $\times$ image bootstrap CIs, Sec. 4). *Real traps*: crossover $[\text{mask} - \text{ERM}]_{\text{out}} - [\text{mask} - \text{ERM}]_{\text{in}}$ between different artifact-bearing images (rebuilt run). *Matched*: the same after 1:1 propensity matching of the in-ROI and out-of-ROI artifact-bearing images (calliper 0.2 SD; Supplement S11). *Transplant*: one real artifact instance pasted inside or outside the ROI of the same artifact-free images (location interaction). *Neutral paste*: neutral tissue pasted through the same masks, positions and blending (paste-edge control). Transplant and matching were designed after the real-trap results and registered before their own results; each design is its own Holm family.

Cohort	Real traps	Matched	Transplant (artifact)	Neutral paste	Artifact – neutral
ISIC hair	+0.147 [+0.109, +0.185]	+0.154 [+0.122, +0.187]	+0.253 [+0.214, +0.293]	+0.204 [+0.164, +0.244]	+0.049 [+0.026, +0.072]
Thyroid	+0.239 [+0.196, +0.280]	+0.238 [+0.186, +0.291]	+0.277 [+0.256, +0.298]	+0.145 [+0.130, +0.161]	+0.132 [+0.116, +0.148]
Ovary	+0.170 [+0.087, +0.253]	+0.163 [+0.049, +0.274]	+0.074 [+0.044, +0.104]	+0.052 [+0.021, +0.084]	+0.022 [−0.002, +0.048]
Capsule	+0.377 [+0.326, +0.425] [†]	infeasible [‡]	+0.497 [+0.427, +0.567]	+0.459 [+0.383, +0.535]	+0.037 [−0.017, +0.093]

[†]19 matched pairs in one label, below the pre-registered minimum of 30. [‡]Descriptive: capsule lesions carrying debris are much larger than those with debris beside them (standardised mean difference 2.6) and matching is infeasible, so this crossover is not by itself evidence about location.

The artifact’s content adds significantly to the paste in dermoscopic hair and thyroid (artifact – neutral: dermoscopic hair +0.049 [+0.026, +0.072], thyroid +0.132 [+0.116, +0.148], ovary +0.022 [−0.002, +0.048], capsule +0.037 [−0.017, +0.093]). By the pre-registered rule the transplant is therefore evidence that *where* a label-correlated pattern sits—not which images carry it—decides what masking removes; it is not evidence about the appearance of any particular artifact, and the in-ROI masking changes it produces (mask – ERM: dermoscopic hair -0.075; thyroid +0.072; ovary -0.010; capsule -0.175) are properties of pasted patterns, not of real artifacts in their natural context. For hair this matters most: a transplanted hair carries the shortcut in its pixels by construction, whereas in the real hair trap the shortcut is carried mostly by correlates of hair (oracle removal of every hair pixel gains only +0.015; Sec. 6), so the transplant shows the location effect for pixel-carried hair, not its size for hair in the wild.

5.4. Covariate-matched traps

As a second, observational check we matched the artifact-bearing images of the two traps 1:1 on a propensity score of lesion or nodule size, artifact amount and acquisition covariates (age, sex, body site, source, diagnosis subtype for dermoscopy; image size and grey level for ultrasound; pre-registered), leaving everything else unchanged, and repeated it with a four times stricter calliper (full balance tables in Supplement S11). dermoscopic hair: 1,829 of 5,438 in-ROI and 8,329 out-of-ROI artifact-bearing images kept per trap, pooled max |SMD| 2.13 $\rightarrow$ 0.11 (within label up to 0.16), crossover +0.147 unmatched, +0.154 [+0.123, +0.186] matched, +0.124 [+0.086, +0.162] with a calliper of 0.05 SD (−16%); thyroid: 316 of 1,150 in-ROI and 326 out-of-ROI artifact-bearing images kept per trap, pooled max |SMD| 0.53 $\rightarrow$ 0.04 (within label up to 0.15), crossover +0.239 unmatched, +0.238 [+0.186, +0.290] matched, +0.218 [+0.165, +0.270]

with a calliper of 0.05 SD (-9%); ovary: 179 of 617 in-ROI and 194 out-of-ROI artifact-bearing images kept per trap, pooled max $|\text{SMD}|$ $0.39 \rightarrow 0.08$ (within label up to 0.14), crossover $+0.170$ unmatched, $+0.163$ $[+0.050, +0.273]$ matched, $+0.152$ $[+0.041, +0.265]$ with a calliper of 0.05 SD (-11%). The effect shrinks by up to 16% under the stricter calliper but every matched crossover excludes zero. Balance is not perfect: within a label some covariates keep standardised differences of 0.13–0.19 after matching (propensity matching balances the score, not every covariate), so residual confounding by these covariates cannot be excluded; the transplant, which holds the images fixed, does not depend on it. For capsule matching left only 19 pairs in one label, below the pre-registered minimum of 30; its real-trap crossover is therefore reported as descriptive, and its location evidence rests on the transplant. A pre-registered doubly robust analysis on the matched samples—per-image placement values of the masked minus the ERM score regressed on the trap indicator, the label and every matching covariate within seed, with a crossed bootstrap—leaves the crossover unchanged: hair $+0.156$ $[+0.127, +0.187]$, thyroid $+0.238$ $[+0.184, +0.291]$, ovary $+0.165$ $[+0.056, +0.273]$ (Holm $p = 0.002$ each); residual imbalance after matching does not carry the effect.

5.5. Masking implementation and the real overlap (pre-registered)

Mean-colour fill is one way to mask. Black fill, a blurred background (16-px Gaussian), a crop to the lesion box and a crop of the masked image all give a positive crossover in all four cohorts (16 of 16, Holm $p \leq 0.010$), from $+0.091$ $[+0.064, +0.117]$ for the dermoscopy box crop to $+0.367$ $[+0.318, +0.416]$ for black fill on capsule frames; implementations that keep part of the out-of-ROI content give weaker effects where that content still carries the artifact (blur for debris and ovarian calipers, the box crop for hair and debris). The two traps are the ends of a continuous dependence: with all artifact-bearing images binned by their real overlap r , the mask gain falls with r in every cohort (slopes: hair -0.235 $[-0.276, -0.194]$, thyroid -0.250 $[-0.303, -0.197]$, capsule -0.489 $[-0.539, -0.438]$, ovary -0.205 $[-0.300, -0.109]$), and for hair it turns from a gain ($+0.124$ $[+0.085, +0.162]$ at $r < 0.1$) into a harm (-0.086 $[-0.119, -0.056]$ at $r \geq 0.75$), crossing zero near $r = 0.40$ (Supplement S16).

5.6. Replication across encoders, training regimes and foundation models

The crossover replicates with MedSigLIP (thyroid $+0.195$ $[+0.153, +0.238]$, capsule $+0.335$ $[+0.286, +0.383]$, ovary $+0.178$ $[+0.109, +0.248]$), a CNN (ConvNeXt-Base: thyroid $+0.275$ $[+0.231, +0.320]$, capsule $+0.320$ $[+0.268, +0.372]$) and DINOv2 at three sizes (positive in all nine cohort-size combinations; Supplement S7), and with end-to-end fine-tuning (crossover of a fine-tuned ResNet-50: hair $+0.143$ $[+0.054, +0.232]$; capsule $+0.606$ $[+0.495, +0.716]$; thyroid ViT-S $+0.122$ $[+0.039, +0.211]$, ResNet-50 $+0.063$ $[-0.022, +0.144]$). The fine-tuned ovary network did not learn the task (ERM clean AUROC 0.60 and 0.50 in Trap A and B, 15 clusters), so it can neither show nor refute a shortcut effect and ovary is not evaluable for this check; the replication holds in all three evaluable cohorts. Without any task training, the zero-shot scores of

vision-language models are already swayed by artifacts, and masking changes them as the law predicts (MedSigLIP crossovers: thyroid $+0.103$ [$+0.064, +0.143$], capsule $+0.045$ [$+0.001, +0.090$], ovary $+0.181$ [$+0.103, +0.261$]; DermLIP on hair $+0.040$ [$+0.021, +0.058$]); MedSigLIP’s zero-shot thyroid diagnosis itself is at chance (clean AUROC 0.45), so we use these scores only to show that the effect is a property of the representation.

The one encoder without a hair gap. With the dermatology foundation model DermLIP and a trained linear head, masking harms at *both* hair locations (Trap A -0.104 [$-0.143, -0.067$], Trap B -0.090 [$-0.135, -0.046$]) and the hair crossover vanishes ($+0.014$ [$-0.037, +0.063$]), although the controlled ruler sweep reverses with DermLIP as with DINOv2 ($+0.161$ at no overlap, -0.263 to -0.309 at 25–100%). Two measurements explain it. First, with DermLIP masking costs disease signal wherever the hair lies (clean AUROC -0.066 [$-0.089, -0.043$] in Trap A and -0.063 [$-0.094, -0.033$] in Trap B): this encoder, trained on whole dermatology images, loses more from a masked background than DINOv2. Second, the pixel part of its hair shortcut is small in both traps (oracle hair inpainting $+0.033$ [$+0.025, +0.042$] and $+0.040$ [$+0.015, +0.064$]), whereas balancing, which also removes correlates, gains $+0.146$ and $+0.110$: masking removes little of the shortcut at either location, so there is no gap to find. The linear-Gaussian account fitted to DermLIP’s clean and correlated AUROC anticipates this (predicted crossover $+0.013$, observed $+0.014$; Sec. 7). The law describes what masking can remove; when the shortcut is carried by correlates and the encoder is fragile to masking, masking removes little at either location.

5.7. Chest radiography: a boundary case

With a synthetic tube swept into the lung fields of NIH pneumothorax radiographs (RAD-DINO), the encoder does not use an extra-pulmonary tube at all (reversed 0.913 versus clean 0.915 at no overlap), so masking has nothing to remove. On real devices (RANZCR-CLiP linked to NIH, four findings; Supplement S3) lung masking helped at both locations with RAD-DINO, the registered primary encoder (no crossover survives Holm correction over the four findings). With MedSigLIP, the registered replication encoder, which does use the out-of-lung tube, the crossover is positive for all four findings ($+0.095$ to $+0.328$, Holm $p < 0.001$) and masking harms inside the lungs for effusion (-0.084 [$-0.125, -0.043$]). The device contrast is confounded, however: images without an in-lung catheter carry more out-of-lung tubes, which lung masking removes. Real chest drains carry the shortcut mostly through correlates (the treated lung; lung masking gains $+0.039$ [$+0.019, +0.058$] with RAD-DINO and $+0.005$ [$-0.013, +0.023$] with DINOv2). With the evidence split by encoder and a confounded contrast, we report chest radiography as the boundary of the law, not as a fifth confirmation.

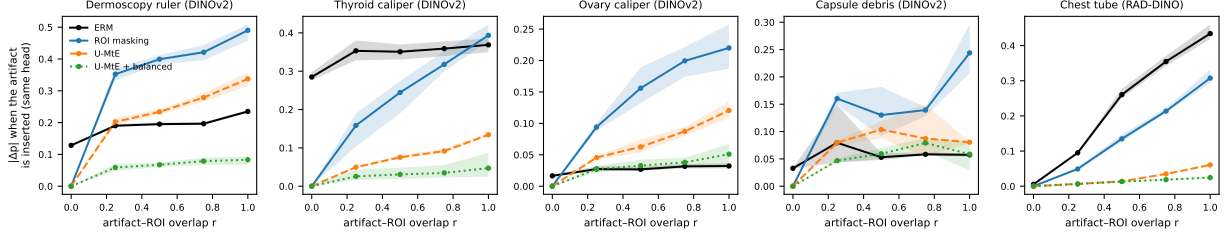


Figure 3: Mechanism: same-head counterfactual sensitivity $|p(x+\text{artifact}) - p(x)|$ versus the artifact’s overlap with the ROI in the five controlled sweeps (mean over seeds; band: range over seeds). Masking removes the out-of-ROI artifact ($r = 0$) but increasingly relies on an in-ROI one; erasure and balancing remove that reliance. Generated by `scripts/analysis/mechanism_figure.py`.

6. Mechanism: masking retains an in-ROI artifact and removes its competitors

Three explanations were tested separately. *Occlusion*: with the ruler present but uncorrelated with the label (50/50), masking changes AUROC by +0.036, +0.041 and +0.039 at 0/50/100% overlap, none of them significantly; covering lesion pixels costs nothing measurable, so the harm requires the correlation. *Lost context*: dilating the mask at 50% overlap *increases* harm ($-0.097 \rightarrow -0.138 \rightarrow -0.153$ for margins 0/10/25 px; neighbouring intervals overlap) as the margin restores ruler pixels (retention $0.50 \rightarrow 0.73 \rightarrow 0.95$); at 100% overlap, where retention is already 1, harm is flat (-0.129 to -0.144) although the ruler’s share of visible pixels falls threefold. *Amplified reliance*: masking doubles the same-head counterfactual sensitivity to the ruler ($|\Delta p|$ $0.235 \rightarrow 0.491$ at 100%). Harm is largest for small lesions (-0.364 [$-0.509, -0.217$] at 100%), where the ruler is a larger share of what survives the mask; for large lesions the effect at 100% is not significant ($+0.063$ [$-0.015, +0.144$]), and at 50% overlap only the small-lesion stratum is significant, so lesion size orders the harm without defining a threshold. Figure 3 shows the same retention mechanism in all five controlled sweeps: with the head held fixed, inserting the artifact changes the masked model’s prediction more the deeper the artifact lies in the ROI. At full overlap this sensitivity exceeds that of ERM for dermoscopic rulers (0.491 vs 0.235; paired difference +0.255 [$+0.221, +0.287$]), ovarian calipers (0.218 vs 0.032; +0.188 [$+0.133, +0.250$]) and capsule debris (0.245 vs 0.057; +0.186 [$+0.130, +0.260$]) and matches it for thyroid calipers (0.393 vs 0.369; +0.025 [$-0.016, +0.065$]). In chest radiography ERM ignores a tube outside the lungs ($|\Delta p| = 0.005$ at $r = 0$), which is why masking has nothing to remove there. Erasing a generic-overlay subspace after masking and group balancing both lower this sensitivity in every sweep (Supplement S8).

What the real-hair shortcut is made of. Removing hair pixels at test time only closes 44–47% of the ERM clean-reversed gap: most of what a frozen encoder exploits in a real hair trap is not the hair pixels but what accompanies hair (hair-bearing images differ in sex, age and body site). This bounds every pixel-level

method, including oracle inpainting and U-MtE, and explains why image-level-label methods, which remove everything correlated with the label, recover more.

7. Why location matters: a linear-Gaussian account

Let a frozen encoder yield a disease block with class-separation signal-to-noise ratio S and an artifact block with SNR A per unit of artifact presence; training uses $P(A=1 | Y=1) = p_1$, $P(A=1 | Y=0) = p_0$ and the reversed test swaps them. For the LDA-optimal linear head the artifact’s effective training SNR is $\tilde{A} = (p_1 - p_0)^2 A / (1 + p_1(1 - p_1)A)$ and

$$\text{AUROC}_{\text{rev}}(S, \tilde{A}) = \Phi\left(\frac{S - \tilde{A}}{\sqrt{2(S + \tilde{A})}}\right), \quad (1)$$

increasing in S and decreasing in $\tilde{A}$ (verified by simulation to within 0.009 AUROC; Fig. 4). Masking maps $(S, \tilde{A}) \mapsto (S_m, \tilde{A}_m)$. Out of the ROI, $\tilde{A}_m = 0$; inside, $\tilde{A}_m \geq \tilde{A}$. Hence (i) with the artifact’s effective strength unchanged ($\tilde{A}_m = \tilde{A}$), in-ROI masking harms exactly when the mask removes disease-informative context ($S_m < S$) and helps only by denoising ($S_m > S$); when the retained artifact also becomes a larger share of what remains ($\tilde{A}_m > \tilde{A}$) masking can harm even though clean performance improves, as for transplanted capsule debris (Sec. 5.3); (ii) for any S_m the benefit of masking is larger out of the ROI than inside it—the crossover is always positive; (iii) erasing a subspace that contains the artifact direction sets $\tilde{A} \rightarrow 0$ but scales S by $1 - \rho^2$, where ρ is its overlap with the disease direction, so erasure beats in-ROI masking unless the artifact resembles the pathology, which disease protection prevents; (iv) group balancing removes the artifact–label association and everything correlated with it, giving $\Phi(\sqrt{S/2})$ irrespective of location; (v) a shortcut carried by correlates that neither the mask nor pixel-level erasure touch can only be removed by label-based reweighting. Each consequence is borne out in Sections 5–9.

Does the account match real encoders? For any test composition $q_y = P(A=1 | Y=y)$ the same head gives $\text{AUROC} = \Phi(\Delta / \sqrt{\sigma_1^2 + \sigma_0^2})$ with $\Delta = S + (p_1 - p_0)(q_1 - q_0)A / (1 + vA)$, $\sigma_y^2 = S + w_a^2(1 + Aq_y(1 - q_y))$, $w_a = \sqrt{A}(p_1 - p_0) / (1 + vA)$ and $v = [p_1(1 - p_1) + p_0(1 - p_0)]/2$. In a pre-registered test we fitted (S, A) for every trained model to its clean and correlated AUROC *only* and compared the implied reversed-test AUROC with the held-out observation; no parameter is shared across models or tuned (Fig. 5). Two reference predictors use the same two inputs: “reversed = clean” (no shortcut harm) and a linear-symmetry heuristic, reversed = 2·clean – correlated. The endpoint on which the account is most informative is the held-out reversed-test AUROC itself: over 136 ERM and masking models (real traps and controlled sweeps) its mean absolute error is 0.052 AUROC ($r = 0.91$), against 0.129 for the symmetry heuristic and 0.237 for “reversed = clean”. Along the controlled overlap sweeps it recovers the sign of the masking gain at 44 of 50 overlap points (symmetry heuristic: 38/50). For the location crossover the gain over the heuristic is small:

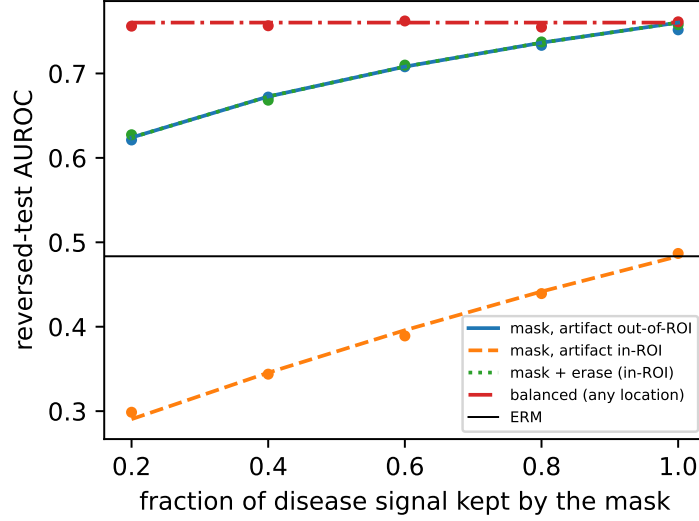


Figure 4: Reversed-test AUROC predicted by the linear-Gaussian model (lines) and simulated with logistic heads (points) as masking removes a growing share of disease signal ($S = 1$, $A = 2$, 90/10 training). Generated by `scripts/analysis/theory_sim.py`.

both follow the observed crossovers closely across nine cohort–backbone pairs (account: mean absolute error 0.015, 9/9 signs; heuristic: error 0.044, 9/9), because a crossover is a difference of masking gains in which most of the symmetry already cancels. We therefore do not use the crossover correlation as evidence for the model. The account also matches end-to-end fine-tuned networks (3/3 crossover signs), which violate its assumptions, and, fitted to DermLIP’s clean and correlated AUROC, it anticipates that encoder’s absence of a hair location gap (+0.013 implied, +0.014 observed; Sec. 5.6). Adding the chest-radiograph cells (device traps and drains, RAD-DINO; 154 models) leaves the error almost unchanged (0.053, $r = 0.88$), but one near-zero device crossover disagrees in sign (consolidation: observed -0.004 , implied +0.008; 12 of 13 signs), which fails our pre-registered all-signs criterion, so we call the model a qualitative account. Its one clear failure is the drain cohort (implied reversed AUROC 0.72–0.78, observed 0.19–0.23): the clean and correlated environments do not expose a shortcut carried by correlates of the drain, regime (v).

Prospective test (pre-registered). The comparison above uses results that existed when the test was designed. We therefore registered the same procedure, unchanged, before the masking-implementation and overlap-bin analyses of Sec. 5.5 were run. On their 84 cells, fitted to each cell’s clean and correlated AUROC only, the account predicted the held-out reversed AUROC with a mean absolute error of 0.011 ($r = 0.996$), against 0.036 for the symmetry heuristic and 0.177 for “reversed = clean”, and met its registered criterion for quantitative prediction. This verdict covers frozen-encoder linear heads on the four real-artifact cohorts; across all cohorts, including the chest-radiograph cells, the account remains a qualitative one.

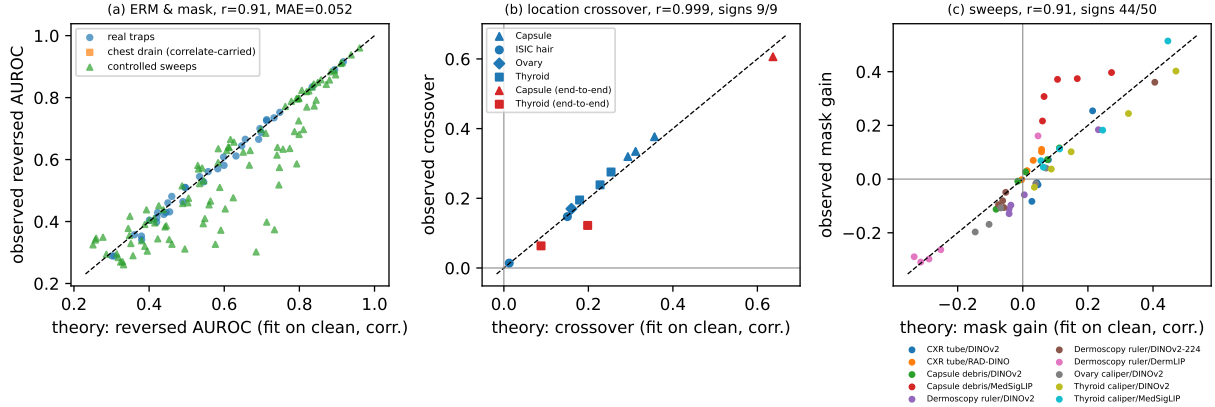


Figure 5: Pre-registered test of the linear-Gaussian account on real encoders. (S, A) are fitted to each model’s clean and correlated AUROC; the reversed-test results are held out. (a) Implied vs observed reversed-test AUROC (ERM and masking). (b) Location crossover per cohort and backbone; red: end-to-end fine-tuned networks, outside the model’s assumptions. (c) Mask – ERM gain along the controlled overlap sweeps. Dashed: identity. Generated by `scripts/analysis/theory_predict.py`.

8. Consequences on unaltered test data

The traps construct an extreme reversal. Without any resampling, in-ROI artifacts are still associated with the label in every cohort (thyroid calipers: 37.6% of benign vs 24.1% of malignant nodules; in-lesion hair: 31% of melanomas vs 20% of benign lesions; capsule debris on the lesion: 23% of polyp-like lesions vs 10% of erosions). We trained on natural training data and tested on untouched data: the official TN3K/TNCD test split, which is patient-disjoint from the training split [12], each ISIC source held out in turn (a real change of hospital and device), and held-out capsule frame blocks.

Discrimination. All analyses in this section use the rebuilt runs (Sec. 10.4). On the shortcut-conflicting pairs masking *lowers* AUROC in 3 of 5 test sets (thyroid -0.095 $[-0.167, -0.023]$; held-out BCN -0.025 $[-0.042, -0.008]$; held-out MSK -0.065 $[-0.111, -0.020]$), raises it for held-out HAM ($+0.050$ $[+0.030, +0.070]$), and does not change it significantly for capsule ($+0.021$ $[-0.003, +0.048]$). Over all test pairs masking changes thyroid AUROC by only -0.004 $[-0.048, +0.041]$: the harm is confined to the cases in which the in-ROI artifact points to the wrong diagnosis, which is exactly where a shortcut is dangerous.

Operating points (pre-registered). Because a clinical decision uses a threshold, we fixed thresholds on each seed’s validation split at five operating points and compared test sensitivity and specificity (Table 3; the bootstrap re-estimates the thresholds in every replicate). On the official thyroid split masking lowers sensitivity at all five operating points, each with a CI excluding zero, for example from 0.699 to 0.424 at the maximal-balanced-accuracy threshold (-0.275 $[-0.435, -0.138]$) and from 0.557 to 0.355 at a validation specificity of 0.80 (-0.202 $[-0.300, -0.100]$). Specificity rises correspondingly, so part of the change is a

shift of the score distribution relative to thresholds learned on validation data (quantified below); the loss is largest where the shortcut conflicts with the label: among the malignant nodules that carry an in-ROI caliper, at a validation specificity of 0.80 sensitivity falls by -0.372 [$-0.499, -0.236$] ($0.549 \rightarrow 0.177$; $n = 78$). *Registration status for this frozen model: a pre-registered test inside a post hoc subgroup.* The subgroup (the 78 malignant nodules of the official test split that carry an in-ROI caliper; the same 78 in every seed) was defined after the first unaltered-data analysis, where its AUROC harm was found; the operating-point test in it was registered before the operating points were computed, and for the fine-tuned model below the whole test was registered before any result. The pre-registered decision rule (loss at the maximal-balanced-accuracy threshold and at ≥ 3 of the four other operating points) is met. On capsule endoscopy, where masking improves AUROC overall, it lowers the sensitivity for erosions carrying debris on the lesion at the two high-sensitivity operating points (Supplement S9).

A fine-tuned model near the benchmark (pre-registered). To test whether these are properties of a weak model, we compared twelve fine-tuning recipes on validation images only, committed the choice (ConvNeXt-T pretrained on ImageNet-22k, learning rate 10^{-4} , 8 epochs, 224 px) and then trained it with five seeds. Its test AUROC is 0.769, 0.004 below the lowest published cross-entropy baseline on this split (0.773; 12) and above the frozen probe (0.735). All four registered tests are supported (Holm): masking lowers the AUROC on shortcut-conflicting pairs by -0.102 [$-0.178, -0.026$]; it lowers sensitivity at all four of OP2–OP5; at OP2 it lowers sensitivity among the 78 malignant nodules with an in-ROI caliper from 0.744 to 0.223 (-0.521 [$-0.642, -0.382$]), a subgroup registered before any result of this model existed; and its trap crossover is $+0.187$ [$+0.111, +0.268$]. Group balancing recovers the conflicting pairs ($+0.088$ [$+0.019, +0.158$] over masking).

Is the loss a threshold artefact? (pre-registered). Re-tuning each model’s threshold on the test set itself—the best that recalibration at a deployment site can achieve—makes the overall sensitivity of the two models equal at a test specificity of 0.80 (-0.017 [$-0.124, +0.070$] for the fine-tuned model), but the conflicting subgroup still loses: from 0.544 to 0.390 (-0.154 [$-0.302, -0.023$]) for the fine-tuned model and from 0.505 to 0.331 (-0.174 [$-0.313, -0.012$]) for the frozen probe (Holm $p = 0.019$ each), while the malignant nodules whose caliper points the right way tend to gain. About 30% of the loss at validation-fixed thresholds therefore cannot be removed by any threshold; the rest follows from transporting thresholds from validation to test data, as deployment does. At a matched specificity of 0.90 and a matched sensitivity of 0.80 the direction is the same, with intervals that include zero (Supplement S16).

New patients at natural prevalence (pre-registered). Trained on all ISIC 2019 images and tested on ISIC 2020 without any resampling, masking lowers the AUROC on shortcut-conflicting pairs: -0.026 [$-0.052, -0.000$] under the registered near-duplicate rule, which proved too loose (it removed most look-alike images of other

Table 3: Masking versus ERM on the unaltered, patient-disjoint official thyroid test split (614 images, 236 malignant) at five operating points fixed on validation data (mean over five seeds; Δ = mask – ERM with 95% crossed seed $\times$ image bootstrap CI, thresholds re-estimated in every replicate). *Conflicting*: the 78 malignant nodules with an in-ROI caliper (the caliper marks benign nodules in this dataset). The subgroup was defined after the first unaltered-data analysis; the operating-point test in it was registered before these numbers were computed.

Threshold on validation	Sens. ERM $\rightarrow$ mask	Δ sensitivity	Sens. conflicting ($n = 78$)	Δ sensitivity, conflicting	Δ specificity
max. balanced accuracy	0.699 $\rightarrow$ 0.424	−0.275 [−0.435, −0.138]	0.703 $\rightarrow$ 0.223	−0.479 [−0.640, −0.313]	+0.202 [+0.115, +0.330]
specificity ≥ 0.80	0.557 $\rightarrow$ 0.355	−0.202 [−0.300, −0.100]	0.549 $\rightarrow$ 0.177	−0.372 [−0.499, −0.236]	+0.117 [+0.059, +0.172]
specificity ≥ 0.90	0.371 $\rightarrow$ 0.219	−0.152 [−0.253, −0.043]	0.336 $\rightarrow$ 0.100	−0.236 [−0.383, −0.089]	+0.075 [+0.029, +0.122]
sensitivity ≥ 0.80	0.753 $\rightarrow$ 0.380	−0.373 [−0.469, −0.270]	0.744 $\rightarrow$ 0.197	−0.546 [−0.671, −0.421]	+0.278 [+0.203, +0.359]
sensitivity ≥ 0.90	0.869 $\rightarrow$ 0.588	−0.281 [−0.391, −0.178]	0.862 $\rightarrow$ 0.400	−0.462 [−0.614, −0.310]	+0.339 [+0.236, +0.425]

patients), and -0.037 [$-0.061, -0.012$] under a label-free calibrated rule registered as a sensitivity analysis after that result (32,952 images). The overall AUROC is nearly unchanged ($+0.016$ [$-0.000, +0.033$]), and group balancing recovers the conflicting pairs ($+0.072$ [$+0.048, +0.097$] over masking). No sensitivity loss at a fixed operating point was detectable, and the cleanest in-lesion comparison (53 hair-free melanomas) was inconclusive (-0.005 [$-0.057, +0.048$]): the external harm combines the removal of hair beside melanomas with the retention of hair on benign lesions (Supplement S16).

Calibration. For the held-out BCN hospital masking worsens both Brier score and expected calibration error (Table 4); for HAM, MSK and capsule it improves every metric, in line with the AUROC results.

9. What to use instead of masking

The location law says what masking cannot do. We therefore asked which remedy removes masking’s in-ROI failure without costing anything where masking already works, and registered the answer’s criterion before testing it.

A class-conditional constraint on the masked features. Let x be the frozen features of the masked image, y the diagnosis and a the image-level artifact label produced by the automatic detectors of Sec. 3, used during training only. On the training images we compute the within-class mean differences $\delta_y = \mu(x \mid a=1, y) - \mu(x \mid a=0, y)$, project the features onto the orthogonal complement of δ_0, δ_1 (or of their average; chosen on validation data) and fit an ordinary logistic head on all training images at their natural weights. Within each class the head can then not score images with and without the artifact differently on average: the shortcut’s contribution, including that of the artifact’s correlates, is removed along the directions the data reveal, while no image is down-weighted. That matters because group balancing loses on unaltered data mainly through the pairs whose two images share the artifact status, i.e. through disease evidence discarded by reweighting (Supplement S17). The constraint is the first-moment, linear case of counterfactual invariance

Table 4: Clinical secondary metrics on the unaltered test sets (Sec. 8): mean (SD) over training seeds. Sensitivity and specificity at the operating point of maximal balanced accuracy fixed on validation data (OP1; other operating points with CIs in Table 3); ECE with 10 equal-width bins; AUPRC baseline = prevalence. Rebuilt run (Sec. 10.4).

Test set	Arm	AUROC	AUPRC	Brier	ECE	Sens	Spec
Thyroid (official split) ($\pi = 0.38$)	ERM	0.735 (0.006)	0.604 (0.006)	0.209 (0.002)	0.088 (0.003)	0.699 (0.045)	0.653 (0.037)
	mask	0.731 (0.005)	0.629 (0.005)	0.215 (0.006)	0.119 (0.025)	0.424 (0.048)	0.856 (0.021)
	U-MtE	0.735 (0.007)	0.630 (0.007)	0.206 (0.003)	0.087 (0.007)	0.419 (0.043)	0.853 (0.020)
	balanced	0.714 (0.005)	0.580 (0.003)	0.220 (0.003)	0.115 (0.006)	0.707 (0.070)	0.612 (0.069)
	U-MtE _{bal}	0.726 (0.006)	0.620 (0.012)	0.207 (0.004)	0.072 (0.024)	0.408 (0.053)	0.851 (0.040)
ISIC, BCN held out ($\pi = 0.23$)	ERM	0.786 (0.003)	0.587 (0.003)	0.238 (0.006)	0.279 (0.011)	0.842 (0.038)	0.521 (0.075)
	mask	0.761 (0.004)	0.564 (0.010)	0.287 (0.007)	0.330 (0.010)	0.842 (0.016)	0.443 (0.033)
	U-MtE	0.767 (0.003)	0.570 (0.007)	0.280 (0.008)	0.327 (0.013)	0.853 (0.011)	0.443 (0.020)
	balanced	0.782 (0.006)	0.587 (0.010)	0.229 (0.006)	0.260 (0.012)	0.835 (0.033)	0.509 (0.071)
	U-MtE _{bal}	0.771 (0.004)	0.567 (0.008)	0.282 (0.010)	0.329 (0.016)	0.869 (0.026)	0.418 (0.058)
ISIC, MSK held out ($\pi = 0.19$)	ERM	0.770 (0.008)	0.477 (0.012)	0.171 (0.007)	0.160 (0.019)	0.600 (0.062)	0.790 (0.048)
	mask	0.785 (0.003)	0.497 (0.006)	0.149 (0.003)	0.120 (0.011)	0.587 (0.060)	0.823 (0.043)
	U-MtE	0.778 (0.002)	0.488 (0.005)	0.154 (0.003)	0.134 (0.008)	0.575 (0.070)	0.818 (0.048)
	balanced	0.767 (0.008)	0.464 (0.014)	0.175 (0.008)	0.165 (0.019)	0.628 (0.043)	0.759 (0.043)
	U-MtE _{bal}	0.770 (0.004)	0.470 (0.006)	0.160 (0.004)	0.141 (0.010)	0.592 (0.050)	0.795 (0.042)
ISIC, HAM held out ($\pi = 0.11$)	ERM	0.752 (0.006)	0.355 (0.006)	0.161 (0.007)	0.233 (0.012)	0.575 (0.084)	0.789 (0.062)
	mask	0.782 (0.004)	0.396 (0.011)	0.138 (0.005)	0.199 (0.011)	0.569 (0.075)	0.831 (0.056)
	U-MtE	0.795 (0.003)	0.402 (0.011)	0.139 (0.006)	0.208 (0.012)	0.606 (0.067)	0.819 (0.046)
	balanced	0.751 (0.016)	0.350 (0.018)	0.154 (0.013)	0.225 (0.014)	0.495 (0.063)	0.845 (0.046)
	U-MtE _{bal}	0.800 (0.008)	0.399 (0.012)	0.135 (0.004)	0.206 (0.015)	0.596 (0.062)	0.828 (0.045)
Capsule (held-out frames) ($\pi = 0.56$)	ERM	0.889 (0.019)	0.897 (0.046)	0.130 (0.016)	0.042 (0.011)	0.770 (0.062)	0.844 (0.055)
	mask	0.967 (0.011)	0.971 (0.014)	0.069 (0.013)	0.029 (0.011)	0.856 (0.034)	0.933 (0.025)
	U-MtE	0.962 (0.009)	0.967 (0.013)	0.081 (0.017)	0.065 (0.045)	0.865 (0.036)	0.904 (0.045)
	balanced	0.881 (0.024)	0.889 (0.053)	0.136 (0.017)	0.051 (0.009)	0.757 (0.086)	0.838 (0.058)
	U-MtE _{bal}	0.959 (0.008)	0.964 (0.012)	0.079 (0.007)	0.045 (0.012)	0.866 (0.050)	0.897 (0.051)

[26]; what is new here is its derivation from the location law and its use after masking. At test time it needs only the masked image, like masking itself.

Registered test against masking. We registered a criterion for *dominating* masking: not worse (margin 0.01 AUROC) on the six unaltered test sets and on the Trap B and clean tests, better on every Trap A and on the conflicting pairs of the unaltered test sets, and no shortcut flipping — 42 components, all of which must hold (Supplement S17). Settings were chosen per training set on validation data only (worst-group validation AUROC, with the same-artifact validation AUROC kept within 0.005 of masking’s, and masking as fallback); confirmation used trap resamples generated after the registration and one cohort (ovary) that never influenced a choice. The constraint met 40 of 42 components (Fig. 6). It is better than masking in every Trap A — thyroid +0.330 [+0.311, +0.351], capsule +0.153 [+0.131, +0.177], dermoscopy hair +0.193 [+0.166, +0.221], held-out ovary +0.118 [+0.089, +0.147] — and on the conflicting pairs of every unaltered test set (thyroid +0.084 [+0.069, +0.101], ISIC 2020 +0.078 [+0.069, +0.088]); it is not worse on the Trap B and clean tests and on four of the six unaltered test sets, and it is inconclusive on the thyroid split (−0.002 [−0.011, +0.007]). Its one loss is ISIC 2020 overall (−0.021 [−0.027, −0.016]): there in-lesion hair still points

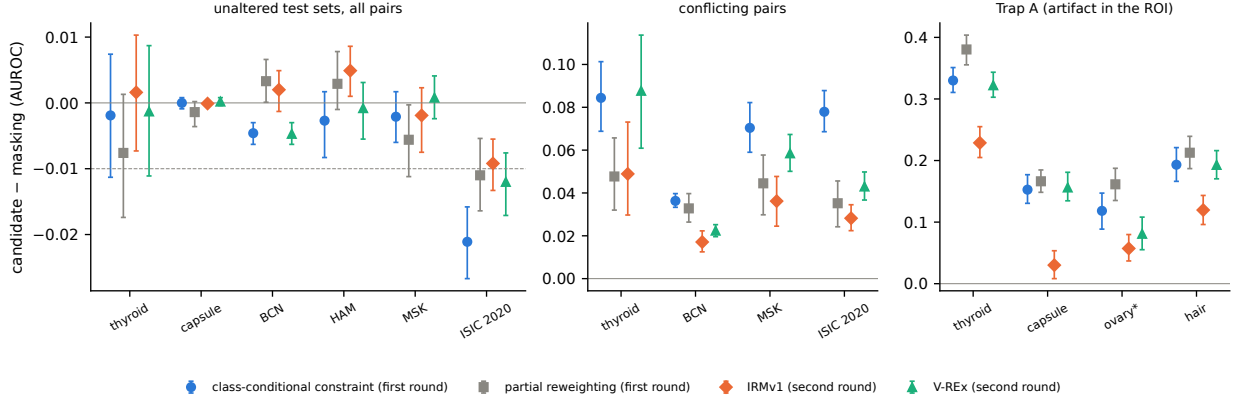


Figure 6: The four candidates closest to dominating masking, each minus masking (AUROC, crossed 95% CIs), on the confirmation data of their registered round (first round: class-conditional constraint, partial reweighting; second round, on new resamples: IRMv1, V-REx). Left: all pairs of the unaltered test sets (dashed: the registered margin -0.01). Middle: shortcut-conflicting pairs. Right: Trap A, the worse of reversed and correlated AUROC; *ovary held out from every choice.

to melanoma for most patients, and removing it costs on those pairs. It covers the three regimes that separated earlier remedies: overlay-like calipers, pathology-like debris and correlate-carried hair.

Nothing does better. A documented search over twelve further ideas on development data only — backdoor adjustment, product of experts and logit adjustment from language and long-tail learning, moment matching, conditional adversarial debiasing, contrastive and invariance methods — and a second registered round of its three best on new data gave the same answer: IRMv1 met 41 of 42 components and lost only on ISIC 2020 overall (-0.009 [$-0.013, -0.005$]), and the constraint, refitted with its frozen recipe, replicated its result (loss on ISIC 2020 only, -0.015 [$-0.024, -0.006$]). Every alternative to masking, including the unmasked model, loses overall on ISIC 2020 (-0.009 to -0.029), mostly on pairs whose images share the artifact status: on new patients at natural prevalence masking is the better disease ranker. No remedy we tested dominates masking.

Without artifact labels. When no artifact label is available, erasing from the frozen features the subspace along which *generic* synthetic overlays move the representation of masked images (mask-then-erase, U-MtE; Supplement S8) removes most of what masking leaves for overlay-like artifacts — thyroid $+0.224$ [$+0.195, +0.253$] over masking with DINOv2, $+0.246$ [$+0.224, +0.269$] with MedSigLIP and $+0.135$ [$+0.102, +0.168$] with a ConvNeXt — but it is worse than masking for debris that resembles the pathology (-0.126 [$-0.146, -0.107$]) and cannot reach a shortcut carried by correlates: removing every hair patch with an oracle mask gains only $+0.015$. On the unaltered test distributions no label-free remedy is uniformly better than masking: annotation-free U-MtE changes AUROC over masking by -0.005 [$-0.011, +0.001$] on capsule and $+0.007$ [$+0.004, +0.010$] for the held-out BCN hospital, its protected variant by -0.023

$[-0.029, -0.017]$ on capsule, and group balancing by -0.086 $[-0.109, -0.065]$ on capsule.

Decision guide. (i) Measure where artifacts sit relative to the ROI; for artifacts outside it, masking is enough. (ii) For artifacts inside it, with image-level artifact labels from an automatic detector, use the class-conditional constraint on the masked features, and expect a small overall cost where the deployment data still reward the shortcut. (iii) Without artifact labels, U-MtE helps for overlay-like artifacts; for correlate-carried shortcuts, which the oracle-removal ceiling on a handful of annotated images diagnoses, labels are needed. (iv) Evaluate any remedy on shortcut-conflicting cases *and* on the unaltered distribution, reporting clean, correlated and reversed performance together, because a reversed-environment gain can be manufactured by removing diagnosis signal (Supplement S8).

10. Robustness and reproducibility

10.1. Robustness to artifact-label definitions

Artifact masks for calipers come from a detector (visual audit: 95% of thyroid detections lie on real calipers; 92–97% of caliper-free images contain no visible caliper). Re-running the thyroid and ovary analyses with large-marker-only (≥ 50 px) and strict-location ($r \geq 0.7$ / $r < 0.05$) definitions leaves the crossover (thyroid $+0.215$ to $+0.262$; ovary $+0.170$ to $+0.195$) and the U-MtE_{protect} gain over masking (thyroid $+0.217$ to $+0.282$; ovary $+0.043$ to $+0.067$) significant in every variant; both grow under stricter definitions, as expected if detector errors dilute rather than create the effects.

10.2. Artifact labels checked without a clinician (pre-registered)

No clinician was available to audit the labels, so we compared them with every independent source we could find, with coverage gates and without the images on which our own labellers were trained (Supplement S16). *Dermoscopy.* Where hair is present our labels place it correctly: our lesion masks agree with the IMA++ multi-annotator masks (241 images; Dice 0.851 [0.830, 0.870]; trap class agreement 84%), and against independent hair masks (485 images) the hair lies on the other side of the lesion boundary in 1 of 264 trap images. But at our threshold (> 30 pixels) the published hair masks mark hair on 93% of the 8,864 images that the expert-reviewed DermArtifactDB labels call hair-free ($\kappa = 0.076$), more often for melanomas. Removing every image an independent source contradicts (42% of each trap) raises the hair crossover from $+0.147$ [$+0.111, +0.185$] to $+0.199$ [$+0.164, +0.233$]: the label noise diluted the effect. *Calipers.* Two independent labellers—an ultrasound annotation detector (BUSClean) and a medical vision-language model (MedGemma, run locally with fixed prompts and never shown our masks)—both pass a pre-registered informativeness rule on thyroid ($\kappa \geq 0.40$) but err in opposite directions, so the registered rule that removes an image whenever either disagrees left too few out-of-ROI caliper images to re-run the thyroid traps. Where the two agree with each other (1,865 of 3,277 images) they agree with our cell for

94%, and without the images both contradict the thyroid crossover is +0.257 [+0.209, +0.304] (exploratory; +0.239 on our labels); the subgroup loss of Sec. 8 holds in the 36 nodules whose in-ROI caliper both confirm. On ovarian screens the annotation detector fires on 90% of caliper-free images and was declared invalid, so the ovary and capsule labels could not be checked with public data.

10.3. Stricter leakage grouping

TN3K/TNCD, MMOTU and SEE-AI publish no patient or video identifiers, so the main analyses group near-duplicates by perceptual hash (and, for capsule, blocks of 100 consecutive frames). As a pre-registered sensitivity analysis we united these groups with connected components of mean-centred DINOv2 embeddings at the most aggressive cosine threshold that keeps every group below 5% of the cohort (thyroid: 3,455 $\rightarrow$ 3,081 groups, $\tau = 0.80$; ovary: 1,059 $\rightarrow$ 969, $\tau = 0.80$; capsule: 137 $\rightarrow$ 135, $\tau = 0.93$) and re-ran all primary contrasts. Eleven of twelve keep their sign and significance: the location crossover is +0.235 [+0.193, +0.275] (thyroid), +0.157 [+0.053, +0.255] (ovary) and +0.353 [+0.296, +0.410] (capsule), and protected U-MtE still beats masking on thyroid (+0.251 [+0.215, +0.284]) and ovary (+0.051 [+0.020, +0.079]). The exception is the small ovary advantage of erasure over augmentation, which loses significance (+0.024 [-0.014, +0.062]). Embedding grouping removes visually near-identical images of one patient or session; it cannot exclude differently looking images of the same patient.

10.4. Rebuild on new hardware

All data were downloaded again and every derived object—caches, perceptual-hash groups, the retrained lesion U-Net (held-out Dice 0.886; HAM10000 agreement 0.895), the caliper scan and the debris probe (IoU 0.625)—was rebuilt on a different GPU generation (pre-registered). Trap membership changes slightly where it depends on a retrained model (hair traps: Trap A 2,655/942 artifact-bearing benign/melanoma images instead of 2,738/989); the thyroid and ovary cells are identical. Every primary crossover reproduces the archived point estimate within the pre-registered tolerance of 0.03 (rebuilt, crossed CIs: dermoscopic hair +0.147 [+0.109, +0.185]; thyroid +0.239 [+0.196, +0.280]; ovary +0.170 [+0.087, +0.253]; capsule +0.377 [+0.326, +0.425]).

11. Discussion and limitations

What the evidence supports. Masking is not a neutral preprocessing step: its benefit depends on where the artifact sits. Four designs agree: real traps, covariate-matched traps (hair, thyroid, ovary), the real-artifact transplant on fixed images, and controlled sweeps, with fine-tuned networks and foundation-model zero-shot scores pointing the same way. It holds for every masking implementation we tried, falls continuously with the real overlap, and grows when images with contradicted labels are removed. The transplant removes the image-population explanation, but its paste-edge control shows that what it isolates is location, not the

artifact’s appearance: neutral tissue pasted through the same masks produces 52–92% of the effect. The evidence that *real* artifacts behave this way in *natural* images therefore comes from the real and matched traps and from the unaltered test data, not from the transplant. The claim is about the *gap*. Whether in-ROI masking helps a little or harms depends on whether the mask also removes disease context, which the linear-Gaussian account makes explicit and which differs between modalities (harm in dermoscopy, small gains in ultrasound and capsule endoscopy).

Why this is not obvious. That a mask cannot remove what it keeps sounds self-evident; four findings show that the consequences are not. First, masking does not merely leave an in-ROI artifact in place: it makes the model rely on it *more*. On the same images, inserting the ruler inside the lesion changes the masked model’s prediction about twice as much as ERM’s (mean $|\Delta p|$ 0.491 versus 0.235 at full overlap; Sec. 6), because masking removes the competing evidence around it. Second, the consequence is clinical and measurable on data as collected: on the patient-disjoint thyroid split masking lowers sensitivity among malignant nodules that carry a caliper, in a subgroup registered in advance for a fine-tuned model near the published benchmark, and part of that loss survives any re-tuning of the threshold; on new patients from ISIC 2020 the harm on conflicting cases is invisible in the overall AUROC. Third, the size of the effect is predictable before it is observed: fitted only to clean and correlated performance, a two-parameter account predicts held-out reversed AUROC with less than half the error of the natural heuristic (0.052 versus 0.129), predicted 84 results registered before they existed (error 0.011), and anticipates the one encoder without a hair gap. Fourth, the neutral-paste control shows that the artifact’s appearance is not what matters: *any* label-correlated pattern inside the ROI survives masking (52–92% of the transplant effect), so the problem cannot be solved by recognising particular artifacts. Together these turn an intuition into a decision rule: where the shortcut sits, and what carries it, decides whether masking helps and what to use instead (Sec. 9).

Why this matters clinically. On the patient-disjoint official thyroid test split the in-ROI caliper is associated with benign nodules; masking keeps it and removes competing evidence, and the malignant nodules that carry a caliper are then missed more often at every validation-fixed operating point. Most of that loss follows from transporting the threshold; re-tuning it on the test set itself removes the overall loss but not the loss in this subgroup. An ROI-masked model can therefore look unchanged by overall AUROC, even after local recalibration, while failing on exactly the cases in which the shortcut points the wrong way. Audits of masked models should stratify performance by artifact presence and location.

Reading reversed-environment gains. A reversed-test gain can be manufactured by removing diagnosis-correlated signal together with the artifact: erasure fitted on unpaired image-level labels and post-hoc prevalence recalibration invert the correlated/reversed ordering. We report reversed, correlated and clean AUROC for every method and summarise robustness by the minimum of correlated and reversed AUROC.

Limitations. (i) *Artifact labels.* Calipers are segmented by a rule-based detector and capsule debris by a patch-token probe (IoU 0.62 against expert masks); hair masks are published semi-automatic masks. No clinician audited them. Independent sources confirm where hair lies, but our hair-presence threshold over-calls hair, with errors that differ by diagnosis; the thyroid caliper labels agree with two independent labellers where those agree with each other, but the registered cleaning rule could not be tested for thyroid, and the ovary and capsule labels remain unchecked (Sec. 10.2). A blinded rating of 240 images by the first author, a non-clinician, is registered and not yet done. Label errors that do not depend on the diagnosis dilute the constructed correlation, and removing contradicted images or using stricter label definitions strengthens rather than weakens the results (Secs. 10, 10.2); the transplant experiment does not depend on the artifact-free group being perfectly clean because the transplanted instance is identical at both locations. (ii) *Patient identifiers.* TN3K/TNCD, MMOTU and SEE-AI publish no patient or video identifiers for the pooled images, so the trap splits group near-duplicates by perceptual hash and frame blocks (a stricter embedding grouping leaves 11 of 12 primary estimates unchanged); the official TN3K/TNCD test split used for the unaltered-data analysis is patient-disjoint. The one open cohort we found with patient identifiers, pathology labels and real calipers (ThyUS2Path; 842 patients) has no nodule masks and uses a scanner interface on which our frozen caliper detector fires on interface elements; external validation on it requires a transferred nodule segmentation and a re-validated detector and is left for future work. For dermoscopy, external validation on new patients was possible and was run under a pre-registered cell-count gate (ISIC 2020, patient-disjoint folds, hair masks from a segmenter trained only on ISIC 2019): the crossover replicates (+0.250 [+0.174, +0.327]), and at natural prevalence masking lowers the AUROC on conflicting pairs (Sec. 8). External validation is limited to dermoscopy, and on ISIC 2020 no operating-point effect was detectable. (iii) *Constructed shifts.* Reversed environments are constructed by subsampling; real deployment shifts are less extreme, which is why we report the unaltered test distributions separately. (iv) *Absolute performance.* Frozen encoders with linear heads are below the best published fine-tuned models on the one cohort with a directly comparable benchmark (official TNCD split: 0.735 AUROC versus 0.773–0.784 for fine-tuned CNNs with cross-entropy and up to 0.799 with curriculum learning, 12). A fine-tuned ConvNeXt-T reaches 0.769, near but below the lowest published baseline, and shows every registered harm (Sec. 8). The law replicates with end-to-end fine-tuned networks in three of four cohorts; on ovarian ultrasound the fine-tuned ResNet-50 barely learns the task and shows no significant crossover (Sec. 5.6). The other cohorts’ binary tasks have no published benchmark on the same split (Supplement S10). (v) *Scope.* Chest radiography is a boundary case: the radiograph encoder ignored extra-pulmonary tubes, the crossover that a second encoder shows rests on a device contrast confounded with other tubes, and real drains carry the shortcut through correlates. The linear-Gaussian account is fitted per model and cannot anticipate a shortcut carried by correlates that the clean and correlated environments do not expose. (vi) *Uncertainty and balance.* The original per-seed bootstrap understated CIs where seeds share test images. All intervals were

therefore recomputed with the crossed bootstrap from regenerated runs (Sec. 4); claims whose interval then included zero were rewritten, and a few secondary analyses from runs that could not be regenerated are given as point estimates only. Matching left within-label standardised differences of up to 0.19 for some covariates. (vii) *Transplant*. Pasted patterns, including neutral tissue, become learnable shortcuts; the transplant establishes location dependence on fixed images, not the size of real artifacts’ effects in their natural context (for hair, whose natural shortcut is carried mostly by correlates, the difference is large). (viii) *Pre-registration*. Every registration is committed before its outcome results (registry with commit hashes and times in Supplement S1); commit times are set by the committing machine, so only the registrations that were pushed to GitHub before their runs (the second review round and every later round) carry an external time stamp, and the 16-test primary family was assembled after the individual results. Two later analyses are sensitivity analyses registered after the result they examine (the calibrated duplicate rule, the cleaner hair groups), and a few are exploratory (the consensus caliper labels). (ix) *Operating points*. Most of the thyroid sensitivity loss at validation-fixed thresholds is threshold transport; the part that survives re-tuning has a wide interval (-0.154 [$-0.302, -0.023$]). (x) *Remedies*. No remedy dominates masking: in two registered rounds the best candidates fail on one cell, ISIC 2020 overall, by 0.01–0.02 AUROC; the dominance tests use frozen DINOv2 features, and the one fine-tuned candidate (a paste-based model) was not a fair test because its pasting could not balance the traps.

12. Conclusion

ROI masking is not a neutral preprocessing step. Its benefit depends on where the artifact sits: it removes shortcuts that live outside the ROI and leaves—or amplifies—those that live inside it. Moving the same real artifact across the ROI boundary of the same images reproduces the effect in all four cohorts, so it is not an artefact of which images happen to carry in-ROI artifacts; it holds across encoder families and model sizes and, where the network learns the task, after fine-tuning; a neutral paste through the same masks behaves similarly, so it is the location of a label-correlated pattern that matters. It follows from a two-parameter model of signal and artifact strength, which predicted new results before they existed. It holds for every masking implementation we tried and is graded in the real overlap. On unaltered clinical data the cost falls on the cases in which the in-ROI artifact points to the wrong diagnosis: it is hidden in the overall AUROC, visible on new patients, and for a fine-tuned thyroid model part of it survives any re-tuning of the decision threshold. No remedy we tested dominates masking, but one comes within a single cell. We recommend that shortcut audits of masked models report artifact location relative to the ROI and evaluate performance on shortcut-conflicting cases alongside clean and correlated performance, and that, where artifacts may lie inside the ROI, a class-conditional constraint on the masked features be used instead of masking alone. Code, pre-registrations, result files and a CLAIM 2024 checklist are public.

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